

Table 1: LoRA and Model hyperparameter settings.

Model	LoRA parameters				Model parameters	
	Target Modules	Rank	Alpa	Dropout	Sliding window	Rope theta
Mistral-7B	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj	64	16	0.1	None	1,000,000
Gemma-7B	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj	64	16	0.1	-	10,000

6 Results

Random Sorting in First Epoch with Subsequent Difficulty-Based Curriculum Learning

To enhance the efficiency of training, we customized the HuggingFace Trainer to perform instruction tuning randomly during the initial epoch. From the second epoch, we systematically arranged the data to align with specific methods for instruction tuning. This approach was designed to assess the impact of structured versus random data presentation on the model’s learning dynamics and overall performance.

In case of Mistral-7B, as shown in Table 2, instruction-tuning on the alpaca dataset did not yield performance improvements over the original model configuration. When comparing various instruction-tuning strategies, the most effective approach involved random data arrangement trained for only one epoch, achieving an accuracy of 60.91%.

However, experiments conducted on the Orca-math dataset revealed that the most effective strategy involved aligning data based on attention mechanisms with three epochs. This approach resulted in the highest mean accuracy of 66.28% as shown in Table 3.

When experimenting with the Slimorca dataset, the most effective training strategy involved organizing the data based on sequence length and limiting the training to two epochs. This method achieved the best performance with an accuracy of 65.08% (Table 4).

For the Gemma-7B model, experiments conducted using the Alpaca dataset indicated that random data arrangement during instruction-tuning yielded the highest average performance, achieving an accuracy of 64.10% (Table 5).

Conversely, results from Table 6 and 7, conducted on the orca-math and slimorca-dedup datasets, indicate that attention-based curriculum

learning was most effective. Specifically, in the orcamath dataset, this method achieved the best performance with an accuracy of 67.54% after two epochs of training. Similarly, for the slimorca-dedup dataset, curriculum learning based on attention mechanisms and extending training to three epochs yielded the highest accuracy of 66.87%.

7 Conclusion and Future Works

While our proposed attention-based method has shown a tendency to improve instruction tuning, the differences observed have not been significantly impactful. Consequently, we plan to upload the results evaluated using the alpaca-eval dataset in version 2 of our project. Currently, our instruction tuning approach involves random training for the first epoch, followed by structured data training in subsequent epochs. Therefore, the results are heavily reliant on the outcomes from the randomly trained model. In the next version, we intend to prepare a pre-arranged dataset and begin the training from the first epoch with this structured data to better assess the effectiveness of this approach.

References

- 2019. Winogrande: An adversarial winograd schema challenge at scale.
- Francisca Adoma Acheampong, Henry Nunoo-Mensah, and Wenyu Chen. 2021. Transformer models for text-based emotion detection: a review of bert-based approaches. *Artificial Intelligence Review*, 54(8):5789–5829.
- Hashir Ali, Ehtesham Hashmi, Sule Yayilgan Yildirim, and Sarang Shaikh. 2024. Analyzing amazon products sentiment: a comparative study of machine and deep learning, and transformer-based techniques. *Electronics*, 13(7):1305.
- Emmanouil Antonios Platanios, Otilia Stretcu, Graham Neubig, Barnabás Póczos, and Tom M Mitchell. 2019. Competence-based curriculum learning for neural machine translation. *arXiv e-prints*, pages arXiv–1903.

Table 2: Results of instruction-tuning the **Mistral-7B** model with the **Alpaca** dataset.

Epochs	Methods	Average (%)	ARC (%) (norm)	HellaSwag (%) (norm)	MMLU (%)	TrustfulQA (%) (mc2)	Winogrande (%)	GSM8K (%) (strict-match)
1	random	60.91	62.12	82.81	61.37	43.74	79.24	36.16
2	random	60.27	63.65	83.33	60.48	42.80	78.45	32.90
	attention	60.72	63.14	83.67	60.90	42.77	79.08	34.72
	loss	59.98	63.40	83.31	60.65	45.79	78.53	28.20
	length	60.58	62.88	83.64	61.01	42.90	79.24	33.81
3	random	59.87	63.14	84.15	59.88	43.26	77.51	31.31
	attention	60.54	63.05	84.06	60.46	42.90	77.98	34.80
	loss	59.84	64.93	84.15	59.54	45.84	77.98	26.61
	length	59.95	63.82	84.21	60.38	43.33	76.48	31.46
Base Model		60.99	61.51	83.47	62.39	42.60	78.05	37.90

Table 3: Results of instruction-tuning the **Mistral-7B** model with the **Orca-math** dataset.

Epochs	Methods	Average (%)	ARC (%) (norm)	HellaSwag (%) (norm)	MMLU (%)	TrustfulQA (%) (mc2)	Winogrande (%)	GSM8K (%) (strict-match)
1	random	65.51	61.01	82.94	61.19	46.36	77.03	64.52
2	random	66.16	60.24	82.92	60.23	47.45	76.87	69.22
	attention	66.20	61.77	83.08	59.66	45.57	77.90	69.22
	loss	65.40	61.18	83.04	60.63	46.73	77.51	63.31
	length	65.89	60.32	82.92	60.33	46.17	77.90	67.70
3	random	65.95	60.49	82.86	59.64	48.69	76.24	67.78
	attention	66.28	60.15	83.35	59.27	47.15	78.37	69.37
	loss	65.58	60.32	82.79	59.60	47.57	75.93	67.25
	length	65.61	60.92	83.16	58.92	47.76	77.03	65.88
Base Model		60.99	61.51	83.47	62.39	42.60	78.05	37.90

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. 2009. Curriculum learning. In *Proceedings of the 26th annual international conference on machine learning*, pages 41–48.

Ernie Chang, Hui-Syuan Yeh, and Vera Demberg. 2021. Does the order of training samples matter? improving neural data-to-text generation with curriculum learning. *arXiv preprint arXiv:2102.03554*.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac’h, Haonan Li, Kyle McDonell, Niklas Muennighoff, Chris Ociepa, Jason Phang, Laria Reynolds, Hailey Schoelkopf, Aviya Skowron, Lintang Sutawika, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. 2023. [A framework for few-shot language model evaluation](#).

Sheng Guo, Weilin Huang, Haozhi Zhang, Chenfan Zhuang, Dengke Dong, Matthew R Scott, and Dinglong Huang. 2018. Curriculumnet: Weakly supervised learning from large-scale web images. In *Proceedings of the European conference on computer vision (ECCV)*, pages 135–150.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. *Proceedings of the International Conference on Learning Representations (ICLR)*.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*.

Shujie Hu, Long Zhou, Shujie Liu, Sanyuan Chen, Hongkun Hao, Jing Pan, Xunying Liu, Jinyu Li, Sunit Sivasankaran, Linquan Liu, et al. 2024. Wavllm: Towards robust and adaptive speech large language model. *arXiv preprint arXiv:2404.00656*.

Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. 2023. Mistral 7b. *arXiv preprint arXiv:2310.06825*.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020.

Table 4: Results of instruction-tuning the **Mistral-7B** model with the **SlimOrca-Dedup** dataset.

Epochs	Methods	Average (%)	ARC (%) (norm)	HellaSwag (%) (norm)	MMLU (%)	TrustfulQA (%) (mc2)	Winogrande (%)	GSM8K (%) (strict-match)
1	random	65.04	63.14	83.09	62.43	49.68	79.64	52.24
2	random	64.53	59.90	83.51	60.95	50.87	78.22	53.75
	attention	64.67	62.03	83.35	60.95	49.58	77.90	54.21
	loss	63.94	63.13	83.27	59.20	51.89	78.61	47.53
	length	65.08	62.12	83.41	61.76	50.31	78.93	53.98
3	random	64.21	62.46	82.95	59.71	50.90	76.40	52.84
	attention	64.22	60.32	83.24	59.60	51.46	77.27	53.45
	loss	65.01	65.35	82.95	60.09	53.63	78.29	49.73
	length	64.40	61.60	83.64	60.20	51.55	76.48	52.92
Base Model		60.99	61.51	83.47	62.39	42.60	78.05	37.90

Table 5: Results of instruction-tuning the **Gemma-7B** model with the **Alpaca** dataset.

Epochs	Methods	Average (%)	ARC (%) (norm)	HellaSwag (%) (norm)	MMLU (%)	TrustfulQA (%) (mc2)	Winogrande (%)	GSM8K (%) (strict-match)
1	random	62.70	61.00	81.95	62.43	47.25	77.58	46.01
2	random	64.10	62.96	82.32	64.20	48.63	77.66	48.82
	attention	63.94	61.86	82.32	63.20	50.40	77.27	48.60
	loss	62.55	61.26	81.97	61.80	50.65	77.19	42.45
	length	62.40	62.37	82.31	61.77	47.85	77.11	42.98
3	random	64.10	63.05	82.28	62.60	51.74	77.03	47.91
	attention	64.06	61.94	82.96	62.95	49.44	77.34	49.73
	loss	62.66	59.98	82.28	62.71	52.61	76.47	41.92
	length	63.92	61.95	82.96	62.95	49.45	77.35	48.89
Base Model		63.30	61.00	82.54	62.82	45.07	77.74	50.64

Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.

Hanyu Lai, Xiao Liu, Iat Long Long, Shuntian Yao, Yuxuan Chen, Pengbo Shen, Hao Yu, Hanchen Zhang, Xiaohan Zhang, Yuxiao Dong, et al. 2024. Autowebglm: Bootstrap and reinforce a large language model-based web navigating agent. *arXiv preprint arXiv:2404.03648*.

Wing Lian, Guan Wang, Bleys Goodson, Eugene Pentland, Austin Cook, Chanvichet Vong, and "Teknum". 2023a. *Slimorca: An open dataset of gpt-4 augmented flan reasoning traces, with verification*.

Wing Lian, Guan Wang, Bleys Goodson, Eugene Pentland, Austin Cook, Chanvichet Vong, "Teknum", and Nathan Hoos. 2023b. *Slimorca dedup: A deduplicated subset of slimorca*.

Stephanie Lin, Jacob Hilton, and Owain Evans. 2021. *Truthfulqa: Measuring how models mimic human falsehoods*.

Shayne Longpre, Le Hou, Tu Vu, Albert Webson, Hyung Won Chung, Yi Tay, Denny Zhou, Quoc V Le, Barret Zoph, Jason Wei, et al. 2023. The flan collection: Designing data and methods for effective instruction tuning. In *International Conference on Machine Learning*, pages 22631–22648. PMLR.

Arindam Mitra, Hamed Khanpour, Corby Rosset, and Ahmed Awadallah. 2024. Orca-math: Unlocking

the potential of slms in grade school math. *arXiv preprint arXiv:2402.14830*.

Subhabrata Mukherjee, Arindam Mitra, Ganesh Jawahar, Sahaj Agarwal, Hamid Palangi, and Ahmed Awadallah. 2023. *Orca: Progressive learning from complex explanation traces of gpt-4*.

Koichi Nagatsuka, Clifford Broni-Bediako, and Masayasu Atsumi. 2023. Length-based curriculum learning for efficient pre-training of language models. *New Generation Computing*, 41(1):109–134.

Narendra Patwardhan, Stefano Marrone, and Carlo Sansone. 2023. Transformers in the real world: A survey on nlp applications. *Information*, 14(4):242.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. Stanford alpaca: An instruction-following llama model. https://github.com/tatsu-lab/stanford_alpaca.

Yi Tay, Shuohang Wang, Luu Anh Tuan, Jie Fu, Minh C Phan, Xingdi Yuan, Jinfeng Rao, Siu Cheung Hui, and Aston Zhang. 2019. Simple and effective curriculum pointer-generator networks for reading comprehension over long narratives. *arXiv preprint arXiv:1905.10847*.

Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivi re, Mihir Sanjay Kale, Juliette Love, et al. 2024. Gemma: Open models